

Alternative Divergence Metrics

This document reports model performance under two alternative divergence-based evaluation metrics: KL divergence and JS divergence. We provide results on three datasets: the World Values Survey (Table D.3), the Big-Five Personality dataset (Table D.4), and the MobLab dataset (Table D.5).

Table D.3. Performance in simulating ideological values (Emancipative Values Index) in the World Values Survey. The JS and KL divergence between the model-simulated $\mathcal{D}_{\text{sim}}$ and the ground-truth human distribution $\mathcal{D}_{\text{obs}}$ is evaluated (the smaller the better). Superscript * marks the augmented baselines. **Bold** highlights the best alignment under each metric.

Method	JS	KL($\mathcal{D}_{\text{obs}} \parallel \mathcal{D}_{\text{sim}}$)	KL($\mathcal{D}_{\text{sim}} \parallel \mathcal{D}_{\text{obs}}$)
1,000 Human Samples	0.02	0.25	0.06
Fixed System Prompt	0.57	20.43	3.36
Persona Hub	0.37	13.48	1.48
OASIS	0.22	5.10	0.87
Modular Pluralism	0.44	16.22	2.06
Persona Hub *	0.22	10.26	0.75
Modular Pluralism *	0.35	11.11	1.37
Ours (EM-based)	0.06	0.71	0.22
Ours (GB-based)	0.06	0.79	0.23

Table D.4. Performance in simulating all five dimensions of the Big-Five Personality dataset. The JS and KL divergence between the model-simulated $\mathcal{D}_{\text{sim}}$ and the ground-truth human distribution $\mathcal{D}_{\text{obs}}$ is evaluated (the smaller the better). Superscript * marks the augmented baselines. **Bold** highlights the best alignment under each metric.

Dimension	Method	JS	KL($\mathcal{D}_{\text{obs}} \parallel \mathcal{D}_{\text{sim}}$)	KL($\mathcal{D}_{\text{sim}} \parallel \mathcal{D}_{\text{obs}}$)
Openness	1,000 Human Samples	0.00	0.08	0.01
	Fixed System Prompt	0.31	10.95	1.14
	Persona Hub	0.36	11.21	1.44
	OASIS	0.20	5.84	0.70
	Modular Pluralism	0.22	7.48	0.79
	Persona Hub *	0.18	8.91	0.59
	Modular Pluralism *	0.18	4.91	0.67
	Ours (EM-based)	0.06	0.38	0.23
	Ours (GB-based)	0.07	2.24	0.29
Conscientiousness	1,000 Human Samples	0.01	0.05	0.02
	Fixed System Prompt	0.39	15.47	1.59
	Persona Hub	0.40	14.78	1.69
	OASIS	0.29	7.93	1.14
	Modular Pluralism	0.28	11.85	1.01
	Persona Hub *	0.24	11.72	0.82
	Modular Pluralism *	0.25	8.72	0.91
	Ours (EM-based)	0.09	2.79	0.31
	Ours (GB-based)	0.07	1.99	0.25
Extraversion	1,000 Human Samples	0.00	0.02	0.02
	Fixed System Prompt	0.30	10.70	1.15
	Persona Hub	0.16	4.73	0.58
	OASIS	0.13	1.47	0.53
	Modular Pluralism	0.30	11.87	1.09
	Persona Hub *	0.04	2.19	0.14
	Modular Pluralism *	0.26	8.06	0.93
	Ours (EM-based)	0.09	1.38	0.33
	Ours (GB-based)	0.05	0.36	0.17
Agreeableness	1,000 Human Samples	0.00	0.05	0.01
	Fixed System Prompt	0.40	16.40	1.63
	Persona Hub	0.30	9.22	1.23
	OASIS	0.42	11.63	1.94
	Modular Pluralism	0.22	8.36	0.76
	Persona Hub *	0.09	5.34	0.29
	Modular Pluralism *	0.17	5.51	0.61
	Ours (EM-based)	0.15	4.01	0.53
	Ours (GB-based)	0.04	0.26	0.17
Neuroticism	1,000 Human Samples	0.00	0.01	0.02
	Fixed System Prompt	0.29	10.77	1.09
	Persona Hub	0.19	5.46	0.70
	OASIS	0.29	10.29	1.11
	Modular Pluralism	0.24	10.21	0.87
	Persona Hub *	0.02	0.96	0.05
	Modular Pluralism *	0.21	9.23	0.71
	Ours (EM-based)	0.03	0.62	0.11
	Ours (GB-based)	0.07	0.49	0.27

Table D.5. Performance in simulating various MobLab games. The JS and KL divergence between the model-simulated $\mathcal{D}_{\text{sim}}$ and the ground-truth human distribution $\mathcal{D}_{\text{obs}}$ is evaluated (the smaller the better). Superscript * marks the augmented baselines. **Bold** highlights the best alignment under each metric.

Game	Method	JS	KL($\mathcal{D}_{\text{obs}} \parallel \mathcal{D}_{\text{sim}}$)	KL($\mathcal{D}_{\text{sim}} \parallel \mathcal{D}_{\text{obs}}$)
Dictator	1,000 Human Samples	0.00	0.01	0.01
	Fixed System Prompt	0.42	16.53	1.60
	Persona Hub	0.42	18.52	1.61
	Modular Pluralism	0.42	18.18	1.72
	Persona Hub *	0.42	18.52	1.61
	Modular Pluralism *	0.41	18.16	1.55
	Ours (EM-based)	0.01	0.04	0.03
	Ours (GB-based)	0.01	0.19	0.03
Proposer	1,000 Human Samples	0.00	0.01	0.01
	Fixed System Prompt	0.22	7.00	0.79
	Persona Hub	0.21	5.58	0.73
	Modular Pluralism	0.20	7.98	0.72
	Persona Hub *	0.10	9.95	0.32
	Modular Pluralism *	0.15	7.80	0.51
	Ours (EM-based)	0.01	0.04	0.03
	Ours (GB-based)	0.01	0.07	0.04
Responder	1,000 Human Samples	0.00	0.01	0.01
	Fixed System Prompt	0.42	14.11	1.72
	Persona Hub	0.24	3.20	0.90
	Modular Pluralism	0.25	12.20	0.86
	Persona Hub *	0.04	2.10	0.12
	Modular Pluralism *	0.25	12.23	0.87
	Ours (EM-based)	0.04	0.24	0.13
	Ours (GB-based)	0.03	1.14	0.11
Investor	1,000 Human Samples	0.00	0.00	0.00
	Fixed System Prompt	0.34	10.10	1.33
	Persona Hub	0.29	7.01	1.08
	Modular Pluralism	0.13	6.37	0.41
	Persona Hub *	0.11	5.71	0.37
	Modular Pluralism *	0.12	6.35	0.41
	Ours (EM-based)	0.01	0.04	0.04
	Ours (GB-based)	0.02	0.12	0.09
Banker	1,000 Human Samples	0.00	0.00	0.00
	Fixed System Prompt	0.35	13.12	1.40
	Persona Hub	0.32	9.89	1.16
	Modular Pluralism	0.19	6.71	0.73
	Persona Hub *	0.18	8.83	0.59
	Modular Pluralism *	0.16	6.62	0.54
	Ours (EM-based)	0.04	0.17	0.14
	Ours (GB-based)	0.02	0.08	0.08
Public Goods	1,000 Human Samples	0.00	0.01	0.01
	Fixed System Prompt	0.36	10.50	1.46
	Persona Hub	0.32	6.17	1.30
	Modular Pluralism	0.23	9.89	0.82
	Persona Hub *	0.09	4.70	0.28
	Modular Pluralism *	0.21	9.78	0.72
	Ours (EM-based)	0.06	0.35	0.21
	Ours (GB-based)	0.07	0.39	0.28
Bomb	1,000 Human Samples	0.00	0.01	0.01
	Fixed System Prompt	0.27	1.58	1.06
	Persona Hub	0.29	1.79	1.11
	Modular Pluralism	0.38	9.10	1.80
	Persona Hub *	0.01	0.05	0.05
	Modular Pluralism *	0.31	8.67	1.46
	Ours (EM-based)	0.06	1.24	0.23
	Ours (GB-based)	0.06	0.27	0.25